

Clément Sebastiao

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Experience

IT University of Copenhagen

Copenhagen, Denmark

PhD Student

Apr 2024–Present

Optimizing urban sustainable mobility networks, part of the Just Streets EU Horizon Program (Grant agreement ID: 101104240). Supervised by Michael Szell and Luca Maria Aiello.

MIT Senseable City Lab

Cambridge, Massachusetts, USA

Visiting Student

Sep 2025–Feb 2026

Gender-Age Mobility Disparities in Tokyo. Supervised by Paolo Santi.

IT University of Copenhagen

Copenhagen, Denmark

Research Assistant

Jan 2025–April 2025

Course Manager and Lecturer of the Master course "Geospatial Data Science".

IT University of Copenhagen

Copenhagen, Denmark

Research Assistant

Jan 2024–Mar 2024

Université de Bourgogne-Franche-Comté

Besançon, France

Intern

Feb 2023–Jul 2023

Mode of Transport and Historical Road Network Growth Relationship, supervised by Claire Lagesse.

ISI

Turin, Italy

Intern

Oct 2022–Jan 2023

Nature and Bicycle Network Relationship, supervised by Rossano Schifanella.

IT University of Copenhagen

Copenhagen, Denmark

Intern

Feb 2022–Jul 2022

Prioritizing Links in Bicycle Network Plans With Subtractive Network Optimization, supervised by Michael Szell.

LIPHY

Grenoble, France

Intern

May 2021–Jul 2021

Collective Motion of Schooling Fish in a Complex Environment, supervised by Aurélie Dupont and Philippe Peyla.

Ecole Normale Supérieure de Lyon

Lyon, France

Intern

Jun 2020–Jul 2020

Deep Learning for Time Series Predictions for FECG Monitoring, supervised by Pierre Borgnat and Patrice Abry.

Université d'Aix-Marseille

Marseille, France

Intern

Jun 2019–Jul 2019

Study of Optically Excited Magnetic States of ZnO by EPR, supervised by Adrien Savoyant.

Education

Ecole Normale Supérieure de Lyon <i>ENS Lyon Diploma,</i>	Lyon, France 2019–2023
Ecole Normale Supérieure de Lyon <i>Master degree in Complex Systems, Physics,</i>	Lyon, France 2020–2022
Claude Bernard Université Lyon 1 <i>Bachelor degree in Physics,</i>	Lyon, France 2017–2020

Languages

French: Native speaker

English: Proficient, A1/A2 *Cambridge Advanced 199/210*

Skills

Research and Productivity Software: QGIS, LaTeX, Visual Studio Code, Zotero, Git.

Python: Network Science (networkx, igraph), Data Science (pandas, numpy, scipy), Geospatial analysis (geopandas, OSMnx, shapely, momepy), data visualization (matplotlib, seaborn, folium).

Other programming languages (decreasing order of proficiency): PostgreSQL, R, Matlab.

Research Interests

Primary: Urban Data Science, Sustainable and Just mobility, Complex Systems

Secondary: Applied Network Science, Data visualization

Peer-reviewed Articles

- [1] Adrien Savoyant, Clément Sebastiao, and Olivier Margeat. Long-Standing Optically Excited Magnetic States in ZnO Nanoparticles. *physica status solidi (RRL) – Rapid Research Letters*, 14(7):2000176, July 2020.
- [2] Clément Sebastiao, Anastassia Vybornova, Ane Rahbek Vierø, Luca Maria Aiello, and Michael Szell. The trade-off between directness and coverage in transport network growth. *Applied Network Science*, March 2026.

Teaching

Courses	
IT University of Copenhagen <i>Geospatial Data Science, 7.5 ECTS</i> Course Manager and Lecturer (100%)	Copenhagen, Denmark <i>Feb 2025 – Jul 2025</i>
Single classes	
IT University of Copenhagen <i>IT and Green Transitions, 7.5 ECTS</i> Urban Data Science and Sustainable Mobility	Copenhagen, Denmark <i>Oct 2024</i>

IT University of Copenhagen
Geospatial Data Science, 7.5 ECTS
Bicycle networks

Copenhagen, Denmark
Jun 2022

Conferences and talks

Network Science Institute

Boston, Massachusetts, USA

Talk

Jan 2026

Leveraging Network Science to Promote a More Just, Sustainable, and Active Mobility

Conference on Complex Systems 2025

Siena, Italy

Talk

Sep 2025

The Trade-Off between Directness and Coverage in Planar Network Growth

Just Streets

Online

Talk

May 2025

Street Space Reallocation: Superblockify

D3A 2.0

Nyborg, Denmark

Poster

Oct 2024

Prioritizing links in bicycle network plans

Conference on Complex Systems 2024

Exeter, UK

Talk

Sep 2024

Prioritizing links in bicycle network plans with greedy network optimization

UrbanSys2024

Exeter, UK

Poster

Sep 2024

Translating active mobility goals into network metrics

MIT Senseable City Lab

Online

Talk

Aug 2024

Pecha Kucha: Bicycle Plans and Superblockify

Nordic Geographers Meeting 2024

Copenhagen, Denmark

Talk

Jun 2024

Prioritizing links in bicycle network plans

PRIMS

Arc-et-Senans, France

Talk

Nov 2023

Morphogénèse urbaine

Rencontres Francophones Transport Mobilité 2023

Dijon, France

Talk

Jun 2023

Inventaire et analyse de motifs viaires pour une compréhension du développement urbain sur le temps long

LIPHY

Grenoble, France

Lightning Talk

Jun 2022

Effets collectifs en environnement complexe chez les poissons

Conference on Complex Systems 2021

Lyon, France

Attending

Oct 2021

Summer schools and advanced courses

Santa Fe Institute

Santa Fe, New Mexico, USA

Complex Systems Summer School

Jun 2025 – Jul 2025

Royal Danish Academy <i>Sustainable Futures IV (PhD Course)</i>	Copenhagen, Denmark <i>Oct 2024 – Jan 2025</i>
Aalborg University <i>The Anatomy of Mobility Injustice (PhD Course)</i>	Aalborg, Denmark <i>Sep 2024</i>
Fondazione Alessandro Volta <i>Advances in Complex Systems Summer School</i>	Como, Italy <i>Jul 2024</i>

Academic services

Reviews	
Scientific Data: 2024	
Nature Cities: 2026	
Organization and committee	
Eurodoc Conference 2025 <i>Volunteer</i>	Copenhagen, Denmark <i>Aug 2025</i>
ITU PhD Club <i>Treasurer, Committee Member</i>	Copenhagen, Denmark <i>Nov 2024–Nov 2026</i>

Academic Awards

UrbanSys2024 <i>Best poster</i>	Exeter, UK <i>Sep 2024</i>
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